GGG-T-563a
JUNE 9, 1958
SUPERSEDING
Int. Fed. Spec. GGG-T-00563 (Navy-Ships)
August 26, 1955 and
Fed. Spec. GGG-T-561
February 6, 1934

FEDERAL SPECIFICATION

TOOLS, BLACKSMITHS': CHISELS, FLATTERS, FULLERS, SET HAMMERS, HARDIES, BOLT HEADING TOOLS, PUNCHES, AND SWAGES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers tools normally used by blacksmiths for cutting, necking, and breaking steel or metals of the same general nature, for smoothing and finishing flat forgings, for necking, grooving and drawing down forgings, for setting down metal to form a square corner, for cutting and trimming forgings, for holding bolts while forming the heads, for punching, clearing and enlarging holes in hot forgings, and for shaping, sizing, and smoothing round forgings.

1.1.1 *Federal specification coverage.*—Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specifications, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.*—The blacksmiths' tools shall be of the following types and classes, as specified (see 6.1):

Type I.—Chisel, handled.
Class 1.—Cold.
Class 2.—Hot.

Type II.—Flatter, square, handled.

Type III.—Fuller.
Class 1.—Bottom, square shank.
Class 2.—Top, handled.

Type IV.—Hammer, set, handled.

Type V.—Hardy.

Type VI.—Heading-tool.

Type VII.—Punch, handled.
Class 1.—Round.
Class 2.—Square.
Class 3.—Backing out.

Type VIII.—Swage.
Class 1.—Bottom, square shank.
Class 2.—Top, handled.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

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Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Governmental:

Department of Commerce Publication:

Simplified Practice Recommendation R17–47.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

Nongovernmental:

Official Classification Committee Publication:

Uniform Freight Classification Rules.

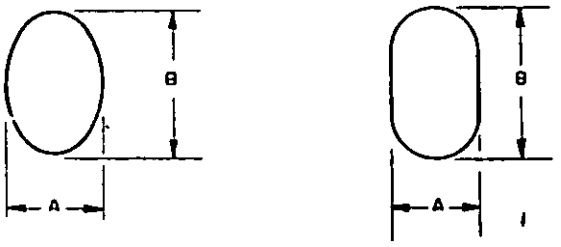
(Application for copies should be addressed to Official Classification Committee, Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations shown herein are descriptive and not restrictive and are included for the convenience of requisitioning and purchasing officers and manufacturers, and are not intended to preclude the purchase of tools which are otherwise in accordance with this specification.

3.2 Material.—The tools, except for type I, class 1 shall be forged from a good quality alloy steel or carbon steel of not less than 0.50 carbon. Type I, class 1 tools shall be forged from an alloy steel. The forging shall be of uniform quality free from burrs and flash marks.

3.3 Eyes.—The handled tools shall be provided with an eye which shall extend through the head, in which the handle shall be fitted. The eyes shall have an inward taper at an angle of approximately 3 degrees to permit proper wedging. The eyes shall be oval in shape in substantial agreement with figure 1 in accordance with Simplified Practice Recommendation R17–47, and conform to sizes specified in the applicable tables.



EYE NUMBER 2 EYE NUMBER 3

FIGURE 1.—Eyes.

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3.4 Handles.—Tool handles shall be made of well-seasoned (not over 12-percent moisture content) straight-grain hickory which shall have not over 22 rings of annual growth per inch of radius and shall be free from any imperfections. The handles shall be smoothly finished and designed to properly fit into the eye of the tool. They shall be well proportioned and facilitate a comfortable handgrip. The handles shall be 16 inches ± 1 inch in length.

3.5 Striking faces.—Tools having striking faces shall have the heads or faces chamfered and slightly rounded.

3.6 Finish.—The tools shall be free from burrs, roughness, and excessive oxide scale. The tools shall have a natural finish, oil coated, painted, or enameled.

3.7 Marking.—Each tool shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.8 Type I, chisel, handled.—Chisels shall be provided with a handle in accordance with 3.4.

3.8.1 *Cutting edge.*—The cutting edge shall be beveled, ground to an included angle of not less than 45 nor more than 60 degrees.

3.8.2 *Hardness.*—The chisels shall show not less than 52 nor more than 58 on a Rockwell C scale when tested as specified in 4.4.1.

3.8.3 *Class 1, cold.*—Class 1 chisels shall be suitable for cutting or necking cold iron and steel, and metals of the same general nature. The chisels shall be provided with either a straight cutting edge or a curved cutting edge of approximately 3¼ inches radius. The chisels shall conform to the dimensions shown in table I for the size specified (see 6.1) and shall be similar to figure 2.

TABLE I.—*Type I, class 1 chisel, handled, cold*

Nominal size	Size of cutting edge C		Length D		Size of stock at eye E		Eye			Weight without handle	
							Number	Figure 1			
								A	B		
	Min.	Max.	Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inches*	*Pounds*	*Pounds*
1¼	1⅛	1⅛	5½	6½	1⅛	1⅛	2 or 3	⅝	⅞	1¾	2¼
1½	1⅜	1⅜	6½	7½	1⅛	1⅛	2 or 3	¾	1	2¾	3¼
1¾	1⅝	1⅝	7½	8½	1⅛	1⅜	2 or 3	1	1¼	4½	5½

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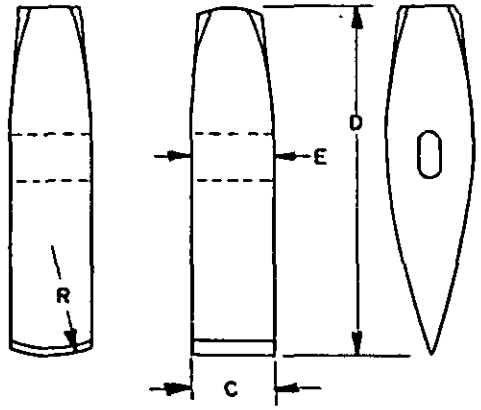


FIGURE 2.—Type I, class 1 chisel, handled, cold.

3.8.4 *Class 2, hot.*—Class 2 chisels shall be suitable for cutting hot iron and steel and other metals of the same general nature. The chisels shall conform to the dimensions shown in table II for the size specified (see 6.1) and shall be similar to figure 3.

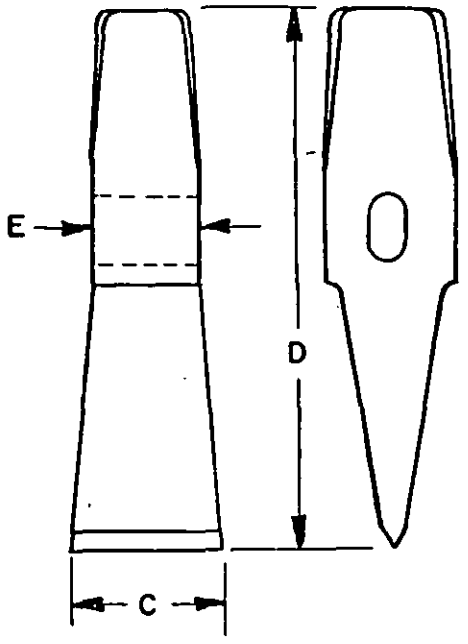


FIGURE 3.—Type I, class 2 chisel, handled, hot.

TABLE II.—*Type I, class 2 chisel, handled, hot*

Nominal size	Size of cutting edge C		Length D		Size of stock at eye E		Eye			Weight without handle	
							Number	Figure 1			
								A	B		
	Min.	Max.	Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inches*	*Pounds*	*Pounds*
1⅝	1 11/32	1 13/32	6⅞	7⅞	1 3/16	1 5/16	2 or 3	5/8	⅞	1¾	2¼
2	1 29/32	2 1/32	7¾	8¾	1 7/16	1 9/16	2 or 3	¾	1	2¾	3½
2¼	2 5/32	2 11/32	8½	10¼	1 11/16	1 13/16	2 or 3	1	1¼	4½	5½

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3.9 Type II flatter, square, handled.—Flatters shall be suitable for smoothing and finishing flat forgings. Each flatter shall be provided with a handle in accordance with 3.4. The face of the flatter shall be square, ground flat, and be uniform in thickness. The flatter shall conform to the dimensions shown in table III for the size specified (see 6.1) and shall be similar to figure 4.

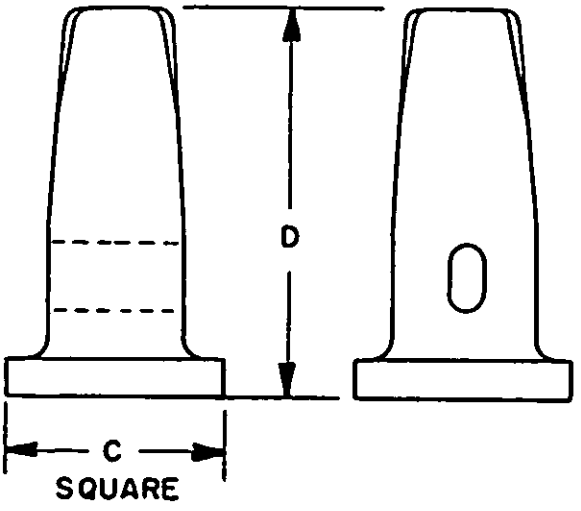


FIGURE 4.—Type II flatter, square, handled.

3.10 Type III fuller. — Fullers shall be suitable for necking, grooving, and drawing down forgings.

3.10.1 *Class 1, bottom, square shank.*— Class 1 fuller shall be provided with a square shank, a suitable shoulder, and a convex radius. The square shank shall be 2⅜ ± ½ inch in length and, unless otherwise specified, 1¼ inches square plus 0 minus 1/16 inch. The face of the shoulder shall be flat and square with the shank. The convex radius shall be accurately and smoothly finished. The fuller shall conform to the dimensions shown in table IV for the size specified (see 6.1) and shall be similar to figure 5.

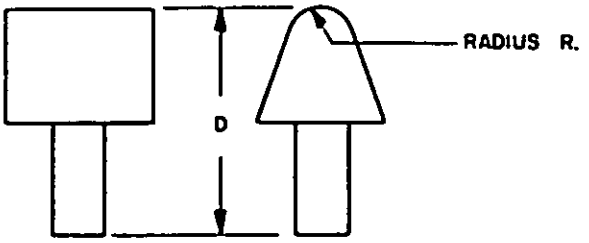


FIGURE 5.—Type III, class 1 fuller, bottom, square shank.

TABLE III.—*Type II flatter, square, handled*

Nominal size	Size of face C		Length D		Eye			Weight without handle	
					Number	Figure 1			
						A	B		
	Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inches	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inches*	*Pounds*	*Pounds*
2	1⅛	2⅛	4	5	2 or 3	⅝	⅛	2	3
2½	2⅛	2⅝	5	6	2 or 3	¾	1	2¾	4¼
3	2⅛	3⅛	5½	6½	2 or 3	¾	1⅛	4¼	7
4	3⅛	4⅛	5½	6½	2 or 3	¾	1⅛	6	8¼

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TABLE IV.—*Type III, class 1 fuller, bottom, square shank*

Nominal size	Radius R			Length D		Weight	
		Min.	Max.	Min.	Max.	Min.	Max.
Inches	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
¼	⅛	7/64	9/64	4	5	1½	2¼
⅜	3/16	11/64	13/64	4	5	1½	2¼
½	¼	15/64	17/64	4	5½	1½	2½
¾	3/8	23/64	25/64	4	5½	2	2¾
1	½	15/32	17/32	4	5½	2¼	3
1¼	5/8	19/32	21/32	4⅛	5½	3	3¼
1½	¾	23/32	25/32	4⅛	5½	3	3½
1¾	7/8	27/32	29/32	4¼	5½	3½	4
2	1	31/32	1 1/32	4¼	5½	3½	4½

3.10.2 *Class 2, top, handled.*—Class 2 fuller shall be provided with a handle in accordance with 3.4. The convex radius shall be accurately and smoothly finished. The fuller shall conform to the dimensions shown in table V for the size specified (see 6.1) and shall be similar to figure 6.

3.11 **Type IV, hammer, set, handled.**—Set hammers shall be suitable for setting down metal to form a square corner and shall be provided with a handle in accordance with 3.4. The tool shall conform to the dimensions shown in table VI for the size specified (see 6.1) and shall be similar to figure 7.

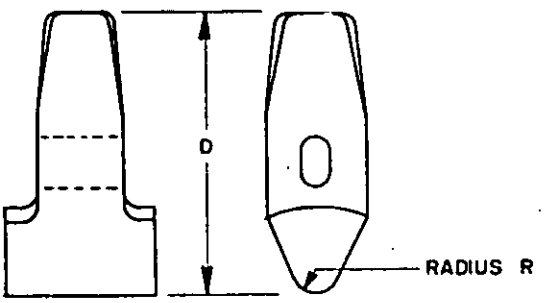


FIGURE 6.—Type III, class 2 fuller, top, handled.

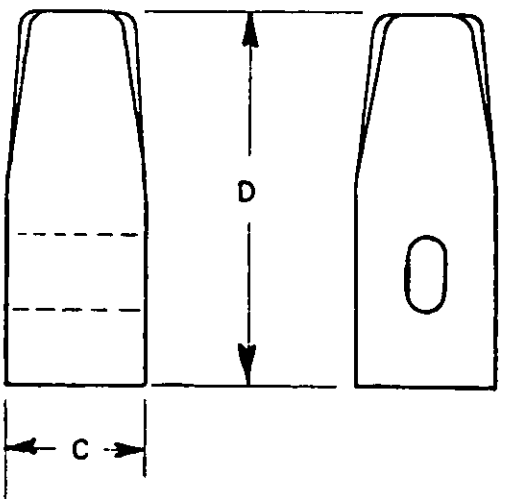


FIGURE 7.—Type IV, hammer, set, handled.

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TABLE V.—*Type III, class 2 fuller, top, handled*

Nominal size	Radius R			Length D		Eye			Weight	
						Number	Figure 1			
		Min.	Max.	Min.	Max.		A Min.	B Min.	Min.	Max.
Inches	*Inches*	*Inch*	*Inches*	*Inches*	*Inches*		*Inch*	*Inch*	*Pounds*	*Pounds*
1¼	1/16	1/64	5/64	4½	5½	2 or 3	5/16	1/16	2	2¼
1⅜	1/16	11/64	13/64	4½	5½	2 or 3	5/16	1/16	2	2¼
1½	1/16	3/64	7/64	4½	5½	2 or 3	5/16	1/16	2	2¼
1¾	1/16	5/64	9/64	4½	6½	2 or 3	5/16	1/16	2½	2¾
1	1/16	3/32	7/32	5¼	6¾	2 or 3	3/8	1	2¾	3
1¼	1/16	7/32	11/32	5¼	7	2 or 3	3/8	1	3¼	4
1½	1/16	9/32	13/32	5¼	7	2 or 3	3/8	1	3¼	4
1¾	1/16	11/32	15/32	5¼	7	2 or 3	3/8	1	4	4½
2	1	13/32	17/32	5¼	7	2 or 3	3/8	1	4¼	4½

TABLE VI.—*Type IV, hammer, set, handled*

Nominal size	Size of face C		Length D		Eye			Weight without handle	
					Number	Figure 1			
	Min.	Max.	Min.	Max.		A Min.	B Min.	Min.	Max.
Inches	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inch*	*Pounds*	*Pounds*
1¼	1 3/16	1 11/16	4	5	2 or 3	5/16	1/16	1½	2
1½	1 3/16	1 13/16	4½	5½	2 or 3	5/16	1/16	2	3¼
1¾	1 3/16	1 17/16	5	6	2 or 3	3/8	1	2¾	4½

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3.12 Type V hardy.—The hardy shall be suitable for cutting and trimming forgings and shall be provided with a square shank and a suitable shoulder. The shank shall be 2⅜ ± ½ inches in length. The face of the shoulder shall be flat and square with the shank. The tool shall have a gradual taper from the cutting edge to the shoulder. The cutting edge shall be ground sharp and flat.

3.12.1 *Hardness.*—The cutting edge of the hardy shall show not less than 52 nor more than 58 on a Rockwell C scale not less than ⅜ inch back from the cutting edge when tested as specified in 4.4.1.

3.12.2 The hardy shall conform to the dimensions shown in table VII for the size specified (see 6.1) and shall be similar to figure 8.

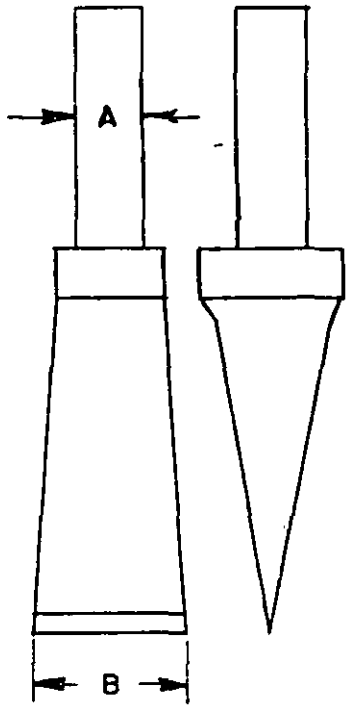


FIGURE 8.—Type V, hardy.

TABLE VII.—*Type V, hardy*

Nominal size of shank A +.000 −.0625	Width of bit B		Weight	
	Min.	Max.	Min.	Max.
Inches	*Inches*	*Inches*	*Pounds*	*Pounds*
½	1 17/64	1 23/64	⅝	1
½	1⅝	1¾	⅞	1¼
¾	1 23/64	1 31/64	1⅛	1½
⅞	1 39/64	2 3/64	1½	2⅛
⅞	2 1/64	2 7/64	1¾	2⅜
1¼	2⅛	2 11/64	2⅛	3
1¼	2⅛	2½	2¼	3¼

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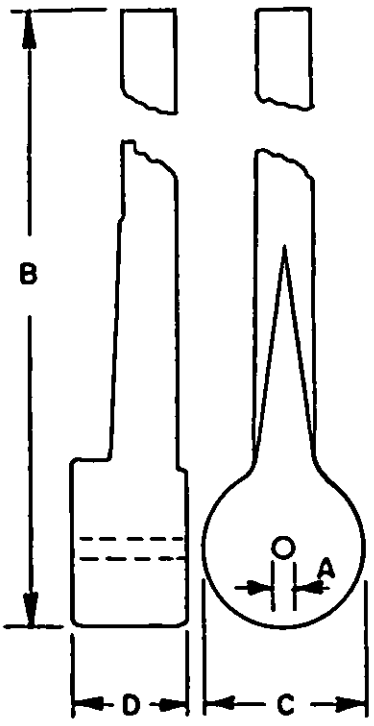


FIGURE 9.—Type VI, heading-tool.

3.13 Type VI, heading-tool.—The heading-tool shall be suitable for holding bolts while forming the heads. The head of the tool shall be round on the sides, the faces flat and uniform in thickness. The hole in the tool shall be true, round, and concentric with the head. The heading-tool shall conform to the dimensions shown in table VIII for the diameter specified and shall be similar to figure 9.

3.14 Type VII, punch, handled.—Punches, except for class 3, shall be suitable for punching, clearing, and enlarging holes in hot forgings. Class 3 punch shall be suitable for backing other punches and removing metal plugs. Each punch shall be provided with a handle in accordance with 3.4. The punch end shall be ground sharp.

3.14.1 *Hardness.* — The punch end and striking end of the carbon steel punches shall show not less than 48 nor more than 55 on a Rockwell C scale. Alloy steel punches shall

TABLE VIII.—*Type VI, heading-tool*

Nominal size of hole A ±1/64 inch	Length B		Diameter C		Thickness D		Weight	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
1/4	13 1/2	16 1/2	2	2 1/2	1 1/8	1 3/8	2 1/2	3 1/2
5/16	13 1/2	16 1/2	2	2 1/2	1 1/8	1 3/8	2 1/2	3 1/2
3/8	13 1/2	16 1/2	2	2 1/2	1 1/8	1 3/8	2 1/2	3 1/2
7/16	13 1/2	16 1/2	2	2 1/2	1 1/8	1 3/8	2 1/2	3 1/2
1/2	14 1/2	18 1/2	2 1/4	2 3/4	1 1/4	1 1/2	3 1/4	4 1/4
9/16	14 1/2	18 1/2	2 1/4	2 3/4	1 1/4	1 1/2	3 1/4	4 1/4
1	14 1/2	18 1/2	2 1/4	2 3/4	1 1/4	1 1/2	3 1/4	4 1/4

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show a Brinell hardness on the punch end of not less than 534 nor more than 578, and the striking end not less than 375 nor more than 401 using 3000 kilogram load, 10 millimeter (mm.) ball or the equivalent on a Rockwell C scale.

3.14.2 *Class 1, round.*—The cutting end of class 1 punch shall be circular in shape, shall conform to the dimensions shown in table IX for the size specified (see 6.1) and shall be similar to figure 10.

3.14.3 *Class 2, square.*—The cutting edge of class 2 punch shall be square in shape, shall conform to the dimensions shown in table X for the size specified (see 6.1) and shall be similar to figure 11.

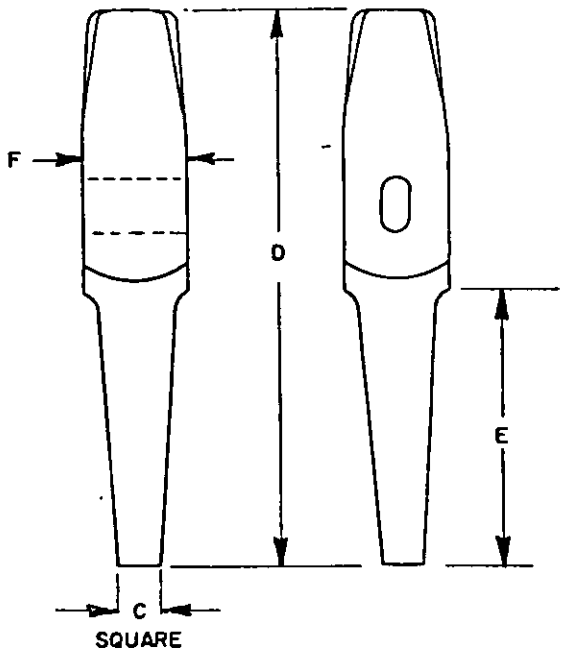


FIGURE 10.—Type VII, class 1 punch, handled, round.

TABLE IX.—*Type VII, class 1 punch, handled, round*

Nominal size C ±1/64 inch	Length overall D		Length of punch E		Size of stock at eye F		Eye			Weight without handle	
							Number	Figure 1			
								A	B		
	Min.	Max.	Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inch*	*Pounds*	*Pounds*
1/4	6 1/2	7 1/2	3	3 1/4	7/8	1 3/8	2 or 3	5/8	7/8	1	1 1/2
3/8	6 1/4	7 3/4	3 1/4	4	7/8	1 3/8	2 or 3	5/8	7/8	1 1/4	1 3/4
1/2	7	8	3 1/4	4	1	1 1/2	2 or 3	3/4	1	1 3/4	2 1/4
5/8	7 1/4	8 1/4	3 1/4	4	1 1/8	1 3/8	2 or 3	3/4	1	2	2 1/2
3/4	7 1/2	8 1/2	3 1/2	4 1/4	1 1/8	1 3/8	2 or 3	3/4	1	2 1/4	2 3/4
7/8	7 3/4	8 3/4	3 1/2	4 1/4	1 1/4	1 3/4	2 or 3	3/4	1	2 3/4	3 1/2
1	8	9	3 1/2	4 1/4	1 1/4	1 3/4	2 or 3	3/4	1	3	4 1/2

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TABLE X.—*Type VII, class 2 punch, handled, square*

Nominal size C ±1/64 inch	Length overall D		Length of punch E		Size of stock at eye F		Eye			Weight without handle	
							Number	Figure 1			
								A	B		
	Min.	Max.	Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inch*	*Pounds*	*Pounds*
1/4	6 1/2	7 3/4	2 3/4	4 1/4	3/8	1 3/16	2 or 3	5/16	3/8	1	1 3/4
5/16	6 1/2	7 3/4	3	4 1/4	3/4	1 3/16	2 or 3	5/16	3/8	1 1/4	2 1/2
1/2	6 5/8	8	3	4 1/4	1	1 1/2	2 or 3	3/4	1	2	2 1/2
5/8	6 3/4	8 1/4	3	4 1/4	1 1/16	1 5/16	2 or 3	3/4	1	2 1/4	2 1/2
3/4	7	8 1/2	3	4 1/4	1 1/16	1 5/16	2 or 3	3/4	1	2 1/2	3
7/8	7	8 3/4	2 7/8	4 1/4	1 1/4	1 3/4	2 or 3	3/4	1	3	3 1/2
1	8	9	2 7/8	4 1/4	1 1/4	1 3/4	2 or 3	3/4	1	3	4

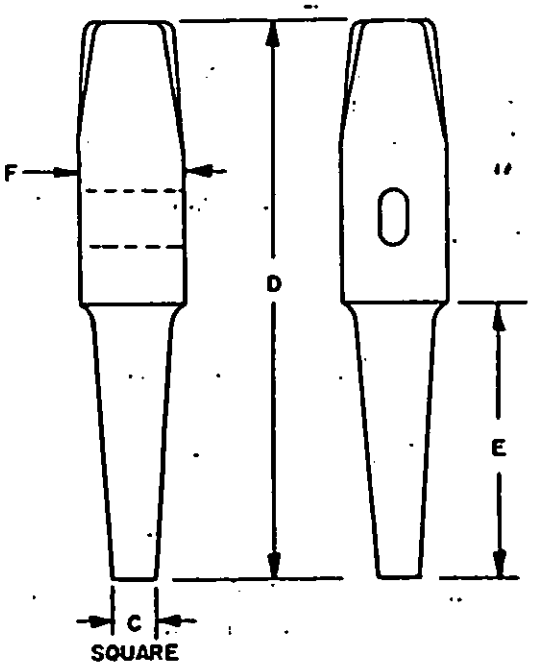


FIGURE 11.—Type VII, class 2 punch, handled, square.

3.14.4 *Class 3, backing out.*—The punch end of class 3 punch shall be circular in shape, conform to the dimensions shown in table XI for the size specified (see 6.1), and shall be similar to figure 12.

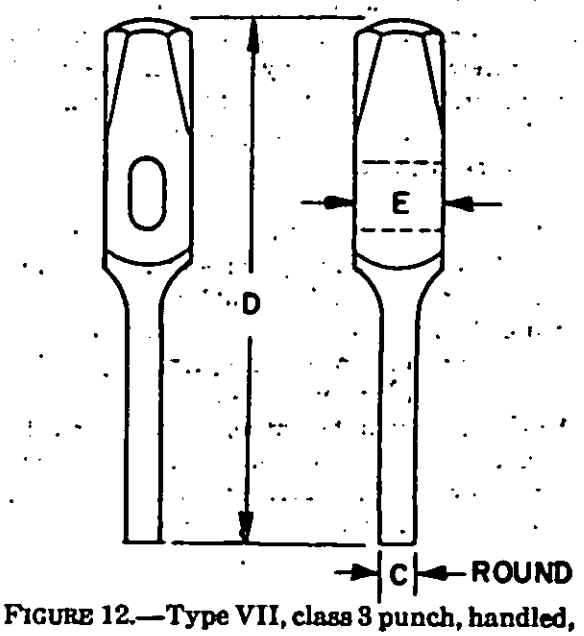


FIGURE 12.—Type VII, class 3 punch, handled, backing out.

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TABLE XI.—*Type VII, class 3 punch, handled, backing out*

Nominal size C ±3/64 inch	Length overall D		Size of stock at eye E		Eye			Weight without handle	
					Number	Figure 1			
						A	B		
	Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inch	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inch*	*Pounds*	*Pounds*
3/8	6 3/4	7 3/4	7/8	1 3/8	2 or 3	5/8	7/8	1 1/8	1 3/4
1/2	7	8	1	1 1/2	2 or 3	3/4	1	1 3/4	2
5/8	7 1/4	8 1/4	1 1/8	1 5/8	2 or 3	3/4	1	2	2 1/4
3/4	7 1/2	8 1/2	1 1/8	1 5/8	2 or 3	3/4	1	2 1/4	3
7/8	7 3/4	8 3/4	1 1/4	1 3/4	2 or 3	3/4	1	2 3/4	3 1/2
1	8	9	1 1/4	1 3/4	2 or 3	3/4	1	3	4

3.15 Type VIII, swage.—Swages shall be suitable for shaping, sizing, and smoothing round forgings. The concave radius shall be accurately and smoothly finished. The depth of the grooves shall be slightly less than the radius "A" dimension, to prevent pinching of the worked metal.

3.15.1 *Class 1, bottom, square shank.*— Class 1 swage shall be provided with a square shank and a suitable shoulder. The square shank shall be 2 3/8 inches ± 1/2 inch in length and, unless otherwise specified, 1 1/4 inches square plus or minus 1/16-inch. The face of the shoulder shall be flat and square with the shank. The tool shall conform to the dimensions shown in table XII for the size specified (see 6.1) and shall be similar to figure 13.

3.15.2 *Class 2, top, handled.* — Class 2 swage shall be provided with a handle in accordance with 3.4. The tool shall conform to the dimensions shown in table XIII for the size specified (see 6.1) and shall be similar to figure 14.

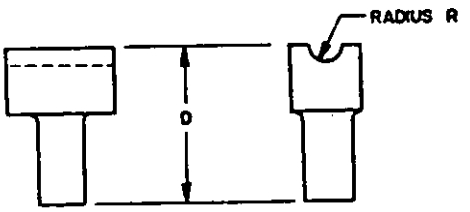


FIGURE 13.—Type VIII, class 1 swage, bottom, square shank.

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TABLE XII.—*Type VIII, class 1 swage, bottom, square shank*

Nominal size	Radius R			Length D		Weight	
		Min.	Max.	Min.	Max.	Min.	Max.
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
¼	5/32	1/16	3/32	3¼	4	2	2¼
5/16	3/16	11/64	1/8	3¼	4¼	2	2¼
⅜	¼	13/64	5/32	3¼	4¼	2	2½
7/16	5/16	15/64	11/32	3½	4½	2¼	2½
½	5/16	17/64	11/32	3½	4½	2¼	2½
9/16	11/32	17/64	13/32	3½	4½	2½	2¾
⅝	13/32	19/64	15/32	3½	4½	2½	3
¾	15/32	23/64	17/32	3¾	5	3	3½
13/16	15/32	25/64	17/32	3¾	5	3	3½
⅞	17/32	27/64	19/32	3¾	5¼	3¼	4½
2	1	31/64	1¼	3¾	5½	3¼	4½
2½	1¼	1⅜	1 11/64	4	5½	4½	5½
3	1½	1 43/64	1 49/64	4¼	6	4½	5½

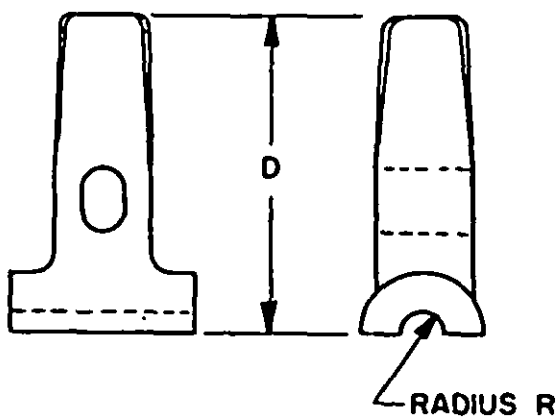


FIGURE 14.—Type VIII, class 2 swage, top, handled.

3.16 **Workmanship.** — Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. Blacksmiths' tools shall be free from any imperfection that may affect their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance inspection.

4.1.1 *Lot.*—All tools of the same type, class, and size presented at one time shall be considered a lot for purposes of acceptance, inspection, and tests.

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TABLE XIII.—*Type VIII, class 2, swage, top, handled*

Nominal size	Radius R			Length D		Eye			Weight without handle	
						Number	Figure 1			
							A	B		
		Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*		*Inch*	*Inch*	*Pounds*	*Pounds*
¼	⅛	7/64	5/32	4	5	2 or 3	⅝	⅞	1	2
⅜	3/16	11/64	7/32	4	5¼	2 or 3	⅝	⅞	1½	2½
½	¼	15/64	9/32	4	5¾	2 or 3	⅝	⅞	1½	2½
⅝	5/16	19/64	11/32	4¼	5¼	2 or 3	⅝	⅞	2	3
¾	⅜	23/64	13/32	4¼	5½	2 or 3	⅝	⅞	2	3¼
⅞	7/16	27/64	15/32	4¼	5½	2 or 3	⅝	⅞	2½	3¼
1	½	15/32	35/64	4¼	5½	2 or 3	¾	1	2½	3¼
1¼	⅝	19/32	43/64	4¼	5¾	2 or 3	¾	1	3	3¼
1½	¾	23/32	51/64	4½	5⅞	2 or 3	¾	1	3¼	4¼
1¾	⅞	27/32	59/64	5	6	2 or 3	¾	1	4	4¼
2	1	31/32	1 3/64	5	6	2 or 3	¾	1	4	4¼
2½	1¼	1 7/32	1 19/64	5	6	2 or 3	¾	1	4½	5½
3	1½	1 15/32	1 35/64	5	6¼	-------	-------	-------	4½	5½

4.1.2 *Sampling for visual and dimensional examination.* — A random sample of tools shall be selected from each lot of material offered for inspection in accordance with Military Standard MIL–STD–105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300 and inspection level I for lots over 300. The acceptable quality level shall be 1.5 percent defective.

4.1.3 *Sampling for lot acceptance tests.*— A random sample of tools shall be selected from each lot offered for tests in accordance with the Appendix to Military Standard MIL–STD–105 at inspection level L–8. The acceptable quality level shall be equal to 4.0 percent defective. However, the sample size shall be the number associated with the letter in the table in the Appendix and the acceptance number shall be zero until the acceptable quality level permits one or greater.

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4.2 Visual and dimensional examination.—Each of the sample tools selected in accordance with 4.1.2 shall be visually and dimensionally examined by the inspector to verify compliance with this specification. Any tool in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective tools in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests.—Each of the sample tools selected in accordance with 4.1.3 shall be subjected to the applicable tests specified in 4.4 to verify compliance with this specification. Any sample tool which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming tools in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Hardness test.*—Tools shall be subjected to a hardness test to determine conformance with 3.8.2, 3.12.1, and 3.14.1.

4.4.2 *Performance test.*

4.4.2.1 *Specimens used for performance test: type I, chisel and type V, hardy.*—A round steel bar showing a hardness of not less than 35 nor more than 40 on a Rockwell C scale shall be used in subjecting the chisels and hardies to the performance test.

4.4.2.2 *Type I, class 1 cold and class 2 hot, handled chisels.*—The sample chisels shall be held in as near a vertical position as possible with the work end resting on the surface of the bar. The chisels shall be struck as squarely as possible a sufficient number of blows to produce an indentation ⅛ inch deep. The class 1 chisel shall make twelve indentations in the test bar. The class 2 chisel shall make six indentations in the test bar. The cutting edge and the head of the chisels shall show no chipping or undue wear.

4.4.2.3 *Type V, hardy.* — Sample hardies shall be set in a vertical position with the shoulder against a solid foundation. The bar shall be held at right angles to the cutting edge of the hardy and struck a sufficient number of blows by a heavy hammer to produce an indentation of ⅛ inch. Six indentations shall be made without the cutting edge of the hardies chipping or showing undue wear.

4.4.2.4 *Type II flatters, type III fullers, type IV hammers, set, type VI heading-tools, type VII punches, and type VIII swages.*—Samples of these tools shall be subjected to a practical test of sufficient duration to determine compliance with the requirements of this specification (see 3.9, 3.10, 3.11, 3.13, 3.14 and 3.15).

4.5 Rejected lots.—Rejected lots may be offered again for inspection and tests provided the contractor has repaired or removed all nonconforming tools. The inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

(Nonmilitary agencies should refer to Federal Standard No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation and packaging.

5.1.1 *Level A.*—Unless otherwise specified (see 6.1), level A cleaning, preservation, and packaging shall be in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Level C cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Level A packing shall be in accordance with Military Specification MIL–P–15424.

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5.2.2 *Level B.*—Level B packing shall be in accordance with Military Specification MIL–P–15424.

5.2.3 *Level C.*—Tools, preserved and packaged as specified in 5.1.1 or 5.1.2 as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Military agencies.* — In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129.

5.3.2 *Nonmilitary agencies.*—Marking for shipment shall be in accordance with the standard marking of Federal Standard No. 123.

6. NOTES

6.1 **Ordering data.** — Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of tools required (see 1.2).
(*c*) Size of tools required (see applicable table).
(*d*) Whether preservation and packaging should be level A or C, and whether packing should be by level A, B, or C (see 5.1).

6.2 Tools covered by this specification are in accordance with Simplified Practice Recommendation R17–47, where applicable.

6.3 **Superseding data.**—Types II, III, IX and X of Federal Specification GGG–T–561 have been deleted. The types formerly covered by Federal Specification GGG–T–561, which are now covered by this specification are as follows:

Specification GGG–T–561	*This specification*
Type I	Type I
Class A	Class 1
Class B	Class 2
Type IV	Type II
Type V	Type III
Class A	Class 1
Class B	Class 2
Type VI	Type IV
Type VII	Type V
Type VIII	Type VI
Type XI	Type VII
Class A	Class 1
Class B	Class 2
Type XII	Type VIII
Class A	Class 1
Class B	Class 2

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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Navy—Sh MC Or Y
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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